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Granular Jamming Gripper for an Ankle Rehabilitation Robot

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Abstract: Rehabilitation exercises are a crucial component in the treatment of ankle injuries. With the increasing demand on physiotherapists, many advances have been made in robot-assisted ankle rehabilitation, which require an attachment mechanism of the foot or leg on the device. However, securely holding the ankle may cause physical discomfort and for the patient to feel constrained to the mechanism. This paper introduces a gripper of the plantar surface of the foot for ankle rehabilitation employing a universal Granular Jamming Gripper (GJG). It consists of a granule filled membrane that allows the patient's foot to be immersed in it when it is filled with air and hardens around the foot when vacuum is applied. The paper also investigates the optimal filling percentage of the membrane, membrane thickness and the effect of granule types of different physical properties (ground coffee, table salt, coffee/salt mixture, and granular sugar) on the gripping ability and overall function of the GJG. Tensile and immersion tests demonstrated the suitability of the GJG for gripping a foot model as well as the increased gripping function with a 0.25 mm membrane filled to 60 % with granulated sugar.

Keywords: Robotics-assisted ankle rehabilitation, granular jamming gripper, ankle rehabilitation gripper.

1 Introduction

The ankle is crucial in maintaining balance, support, and locomotion. Yet, ankle injuries are the most common musculoskeletal injuries. Furthermore, neurological injuries have been listed as one of the main reasons for ankle disability [1]. According to a report from the American Heart Association, around 795,000 individuals experience strokes annually in the United States, with stroke-related complications such as drop foot contributing to a substantial number of permanent disabilities worldwide [2]. Therefore, addressing ankle dysfunction requires addressing damage to the central nervous system through stimulation for reorganization and compensation to recover proprioception. That makes rehabilitation an essential component of treatment. Rehabilitation is performed by a physiotherapist directly on the patient and by instructing active exercises to be performed by the patient, with aids such as balance boards and terra-bands [3]. However, one-on-one traditional ankle rehabilitation training poses challenges of time, resources, and the provision of sufficient frequency and intensity of training. Previous studies showed that without sufficient rehabilitation, 44 % of people will have future problems of ambulation, re-injury prevalence, and approximately 38 % of people will have recurrent activity limitation [4].

The integration of robotics technology holds the potential to effect a much-needed transformation, helping in patient diagnosis, therapy customization, and maintaining patient records. Among the developments in ankle rehabilitation robotics are platform-based devices that specifically target the improvement of ankle performance by enabling actuation across all three significant degrees of freedom [2].

The limited to absent intervention of the therapists in robotic ankle rehabilitation requires fixing the foot or leg to the device. However, ankle rehabilitation robots were found to be undesirably inspired by industrial robot designs [5]. The commonly used mechanisms for attaching the foot are belts, straps and elastic bands which constrain the leg or foot and can become uncomfortable over long rehabilitation periods [2].

This paper presents a universal Granular Jamming Gripper (GJG) for ankle rehabilitation, that attaches the plantar surface of the foot, molding to its geometry and allowing the patient to release their foot easily when the gripper is turned off. Furthermore, it investigates an optimal composition of the granule filled membrane of the GJG through the measurement of the maximal gripping forces and subjective evaluation of user-friendliness and manufacturability criteria. The explored parameters are filling percentage of the membrane, membrane thickness granule type.

2 System design

The GJG is designed to allow the immersion of the plantar surface of the foot in a granule-filled flexible membrane,

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followed by introducing vacuum in the membrane thus gripping the foot by hardening the granular material around it [6]. A vacuum pump (Thermo Fisher Scientific Inc., MA, USA) boasting a flow rate of 32 l/min was connected through an air filter to the membrane.
According to BROWN et al., the GJG requires a vacuum of 1/4 atm (≈ 0.253 bar). Therefore, the air and vacuum flow are directed with two 24 V solenoid valves (Heschen Electric Co.Ltd, Foshan, China) operating in the pressure range 0-0.7 MPa (0-7 bar). Their control was realised with an Arduino Uno (Arduino S.r.l., Monza, IT) (see Figure 1).

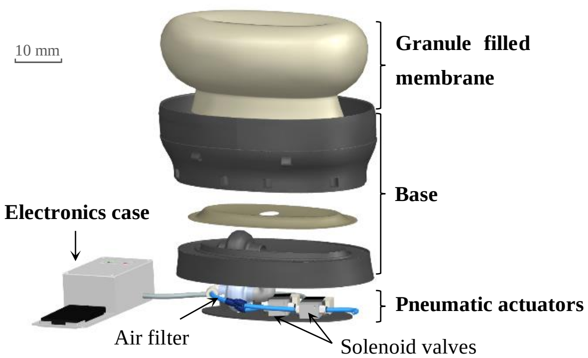

Figure 1: Main components of the GJG

The geometry of the membrane is designed to enhance the holding force of the GJG, which the result of three separate mechanisms, friction, suction and interlocking. The latter significantly is increased when at least half of a convex object is enveloped by the gripper. For the plantar surface of the foot, the membrane's walls need to surpass the foot's widest circumference, defined by contact points between the foot and tangent lines parallel to the immersion direction. Thus, the granular mass needs to form an irregular half toroid around the foot, achieved by shaping the membrane as an irregular ellipsoid with a depression in the centre of the top surface.

The membrane is constructed by brushing "Formalate" liquid latex (CREARTEC trend-design-gmbh, Lindenberg, DE) onto a 3D-printed mold mounted on a rotating axis (see Figure 2) [7]. Latex provides a flexible membrane, allowing granular mass deformation around the foot. Its surface offered high static friction and airtight sealing, crucial for regulating airflow, maintaining vacuum inside the membrane, and enabling the suction effect [8]. The membranes that this set-up allowed to produce without tears or irregularities are 0.15 mm, 0.2 mm, and 0.25 mm thick.

The gripper's base comprises two parts allowing disassembly for membrane and granule replacement. The bottom part houses pneumatic components, including an air filter isolating granules from air lines, attached to the membrane's base and a 13 mm silicone tube. Two 24 V normally closed solenoid valves connected to the filter via 6

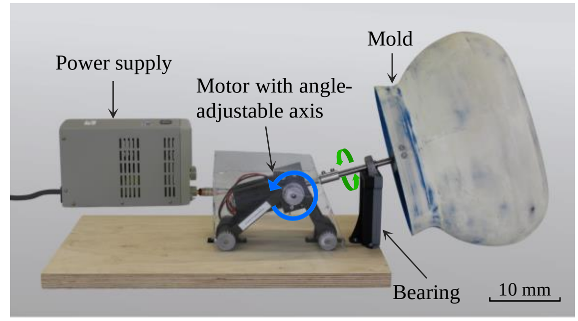

Figure 2: Molding set-up for the latex membrane. Green: Direction of mold rotation, blue: direction of motor adjustment.

mm polyurethane tubes, controlling vacuum and air ports. The top part is a rigid collar surrounding the membrane and complementing its geometry, thus guiding its deformation around the foot. Between the two parts of the base a latex sheet is laid acting as a base for the membrane that connects to the air filter. It bridges the gap from the wide opening of the membrane that was necessary for the ability to peel it off the mold. The airtight sealing of the membrane and latex sheet is affected by entrapping them between the two parts of the base (see Figure 3). The base was made using a Prusa i3 MK3S 3D-printer (Prusa Research s.r.o., CZ) and polylactic acid filament given its irregular shape and the intricate sealing structure.

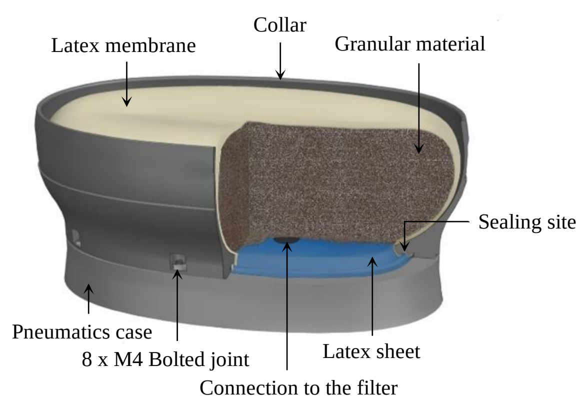

Figure 3: GJG membrane and base with a cross-section showing the airtight sealing of the membrane and latex sheet.

3 Membrane and granules selection

The parameters filling percentage, membrane thickness, and granule type were defined by comparing their effects on gripping force, ease of insertion, durability, device mass, and reproducibility, on each parameter following guideline VDI-2225.

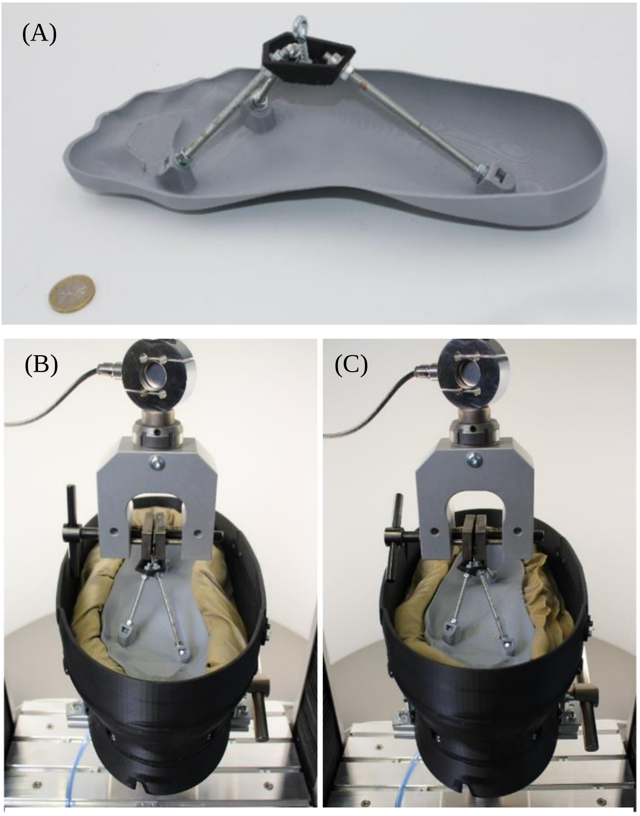

Figure 4: (A): model of the 95th percentile German foot sole, (B): Foot model enveloped by GJG's membrane on the UTM, (C): GJG hardened around the foot model during pull test.

The criteria *gripping force*, *ease of insertion*, *membrane durability*, and *device mass* were each scored from 0 to 4 for each parameter, based on subjective grades. Ease of insertion was subjectively compared and rated from "insertion not possible" to "minimal resistance" using a 3D-printed model of the 95th percentile German male's foot sole (see Figure 4), manually inserted until fully enveloped by the membrane.

A pull-test on a Quasar 25 Universal Testing Machine (UTM) (CESARE GALDABINI S.p.A., Cardano al Campo, IT) determined the maximal gripping force (MGF) for each combination of options from each parameter, with scores ranging from "lower than 80 N" to "higher than 170 N". The manually inserted foot model was attached to the mobile beam of the UTM while the GJG was fixed to the base of the UTM. The foot model was pulled upwards from the operating GJG and the gripping force was recorded.

The criteria *membrane durability* and *device mass* were evaluated on scales ranging from "non-functional" to "durable", and from "heavier than 4 kg" to "lighter than 2 kg", respectively.

Tests comparing filling percentages of the membrane with granules used a 0.25 mm membrane filled with ground coffee to 50%, 60%, and 70%, as volumes outside this range did not sufficiently immerse the foot model. For the filling of the membrane, four types of granules and mixtures were chosen: ground coffee, table salt, a 2:1 mixture of ground coffee and table salt (C+S), and granulated sugar. These were tested with a 60% filled 0.25 mm membrane. Additionally, the effect of membrane thickness was investigated using 0.15 mm, 0.2 mm, and 0.25 mm membranes, filled with the coffee/salt mixture at 60%.

4 Results

The foot was immersed manually in the membrane such that the widest horizontal circumference of the foot is enveloped with granules in order to maximize the gripping force. The granules inside the latex membrane flowed to conform to the shape of the foot. When the vacuum input connected to the membrane is and the air input is blocked, the granules jam together and form a rigid mass that conforms to the foot's geometry. Gripping force was thus generated through a combination of interlocking between complementary geometries and the surface properties of latex. When the air input is connected to the membrane is and the vacuum input is blocked, the granules return to a fluid-like state immediately allowing the foot to be released. This function was realised by all the investigated choices of granule types, membrane thicknesses and filling percentages. However, the gripping forces varied depending on the combination, the highest being for the 0.25 mm membrane filled to 60 % with granulated sugar (see Table 1).

Table 1: MGF across options of filling percentage, granule type and membrane thickness; (A): with ground coffee granules, 0.25 mm membrane**,** (B): with a 0.25 mm membrane, 60% filling, (C): with 60% filling, coffee + salt granules.

Filling percentage(A)		**Granule type(B)**		**Membrane thickness(C)**	
Percentage (%)	MGF (N)	Type	MGF (N)	Thickness	MGF (N)
50	65	Coffee	66	0.15	137
60	90	Salt	159	0.20	154
70	133	Coffee + salt	77	0.25	70
		Sugar	227		

Table 2 shows a ranking of the choices following their evaluation according to the weighted criteria gripping force, ease of insertion, durability, mass of the device and reproducibility.

Table 2: Comparison of filling percentages, granule types, and membrane thicknesses based on weighted criteria

	Filling percentage				Granule type					Membrane thickness			
	Weighting	50 %	60 %	70 %	Weighting	Coffee	C+S	Salt	Sugar	Weighting	0.15	0.2	0.25
Gripping force	4	2	2	3	4	1	1	3	4	3	2	3	3
Ease of immersion	4	3	4	3	4	1	4	2	3	3	3	4	4
Durability	1	4	4	4	2	4	3	3	2	4	4	2	1
Mass of the device	3	4	3	2	1	4	3	1	1	0	4	4	4
Reproducibility	0	4	4	4	0	4	4	4	4	4	4	2	1
Weighted sum		**36**	**37**	**34**		**20**	**29**	**27**	**33**		**50**	**45**	**29**
Score (%)		**75**	**77.1**	**70.8**		**45.5**	**65.9**	**61.4**	**75**		**89.3**	**80.4**	**51.8**
Ranking		**2**	**1**	**3**		**4**	**2**	**3**	**1**		**1**	**2**	**3**

5 Discussion

The design and testing of the GJG are demonstrated, allowing for the assessment of its suitability in gripping the plantar surface of the foot. Moreover, the evaluation of membrane thicknesses, filling percentages and different granular materials determined that 0.25mm-thick membrane, 60% filling, and granulated sugar, respectively, are optimal for this demonstrator. However, the system faces challenges that require optimization. Firstly, while sugar as the granule type achieved the highest rank for the demonstrator, it is not optimal in all criteria, particularly due to its sensitivity to temperature and friction, which alter the flow of the granules [9]. Additionally, difficulties were identified in the insertion of the foot into the membrane. Improvement in this aspect is crucial, especially considering that the gripper targets patients with neurological impairments who may lack the necessary force for adequate foot immersion. The choice of granules affected the ease of insertion, with larger and rougher granules reducing the required insertion force. Hence, the need for a supplementary mechanism, e.g. implementing vibration.

Further developments include measuring the pressure of the GJG on the foot and further investigating the above-mentioned parameters to ensure the highest gripping force within the margins of safety. Moreover, an implementation of a rechargeable pressure tank is proposed to produce vacuum in place of the vacuum pump due to its loud noise, non-portability, and high energy consumption. In later stages, efforts can be directed towards integrating the GJG in ankle rehabilitation devices.

Author Statement

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References

[1] X. Zeng, G. Zhu, M. Zhang, and S. Q. Xie, "Reviewing Clinical Effectiveness of Active Training Strategies of Platform-Based Ankle Rehabilitation Robots," *Journal of Healthcare Engineering,* vol. 2018, p. 2858294, 2018/02/20 2018, doi: 10.1155/2018/2858294.

[2] M. Dong *et al.*, "State of the art in parallel ankle rehabilitation robot: a systematic review," *Journal of NeuroEngineering and Rehabilitation,* vol. 18, no. 1, pp. 1-15, 2021.

[3] S. T. Hawson, "Physical Therapy and Rehabilitation of the Foot and Ankle in the Athlete," *Clinics in Podiatric Medicine and Surgery,* vol. 28, no. 1, pp. 189-201, 2011.

[4] B. L. Braun, "Effects of ankle sprain in a general clinic population 6 to 18 months after medical evaluation," *Archives of family medicine,* vol. 8, no. 2, p. 143, 1999.

[5] Y. M. Khalid, D. Gouwanda, and S. Parasuraman, "A review on the mechanical design elements of ankle rehabilitation robot," *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine,* vol. 229, no. 6, pp. 452-463, 2015.

[6] F. N. Nogueira, T. Mendonça, and P. Rocha, "Positive state observer for the automatic control of the depth of anesthesia-Clinical results," (in eng), *Comput Methods Programs Biomed,* vol. 171, pp. 99-108, Apr 2019, doi: 10.1016/j.cmpb.2016.08.019.

[7] M. Pushpakath and M. H. Ang Jr, "Design of a Liquid Jamming Gripper," *Designs,* vol. 7, no. 2, p. 44, 2023.

[8] J. R. Amend, E. Brown, N. Rodenberg, H. M. Jaeger, and H. Lipson, "A Positive Pressure Universal Gripper Based on the Jamming of Granular Material," *IEEE Transactions on Robotics,* vol. 28, no. 2, pp. 341-350, 2012, doi: 10.1109/TRO.2011.2171093.

[9] L. C. d. Santos, R. Condotta, and M. d. C. Ferreira, "Flow properties of coarse and fine sugar powders," *Journal of Food Process Engineering,* vol. 41, no. 2, p. e12648, 2018, doi: https://doi.org/10.1111/jfpe.12648.